



CL-2001

Data Structures

Lab Task # 2

Objectives:

- ADT Implementation of Linked List
- Operations on Linked List
- Solving real world problems using lists

Note: Carefully read the following instructions

1. Use proper font family (Calibri or Times New Roman) and font size of title (16 points), heading (14 points), sub heading (12 points) and normal text (10 points).
2. First think about problem statement and then write/draw your logic on paper.
3. **Microsoft Visual Studio** should be used to make C++ programs. Programs made with any other software would not be accepted.
4. For each task, create a new C++ program with the naming convention as follows: **TASK-NO**
5. Mention what is happening in each line of code using comments.
6. Write all codes one by one with proper numbering and also paste screen shot of each problem using **snipping tool**(default screen capture software in windows) on **Microsoft word file**.
7. You are to submit a total of 6 files. 5 .cpp files of your code and 1 docx file containing all codes together and corresponding screenshots. Please submit your file with this naming convention **ROLLNO_SECTION_GROUPNO_LABNO**.
8. Do not copy from any source otherwise you will be penalized with ZERO grades.
9. Submit your lab on **Google Classroom**.

**Problem 1: Search an Element in a Linked List****10**

You are to write a function that searches for a key in a linked list given its head pointer. Using recursion.

Problem 2: Find the Second Maximum Number in a linked list**10**

You are to write a function that returns the second maximum number in list.

Problem 3: Linked list Re-adjustment**20**

You are to write a function that first check if linked list is even-sized or odd-sized. If size is even then add an element to first node or last node. If size of the linked list is odd then it should insert the node in the middle.

And returns the list after the operation.

Problem 4: Singly Linked List**30**

Implement a singly linked list class with the following problem:

You are building a music player application that stores a playlist of songs. Each song has a title and an artist. You need to design the data structure and provide operations to add a song to the playlist, remove a song from the playlist, and display the current playlist. Implement following operations

1. Add song/ songs anywhere in a playlist.
2. Remove song/ songs anywhere from a play list.
3. Display Playlist of all added songs

Problem 5: Singly Linked List**30**

Implement the TextEditor using a singly linked list to store the document. The class should provide the following operations:

1. insert_line(line_number, text): Inserts a new line with the given text at the specified line number. If the line number is out of range, it should display an error message.
2. delete_line(line_number): Deletes the line at the specified line number. If the line number is out of range, it should display an error message.
3. edit_line(line_number, text): Edits the line at the specified line number with the given text. If the line number is out of range, it should display an error message.
4. get_line(line_number): Returns the text of the line at the specified line number. If the line number is out of range, it should return None.
5. display_document(): Displays the current document, i.e., all the lines of text in order.